

Experiment No. 6

Title

Breadth First Search (BFS) Using Artificial Intelligence

Aim

To develop a Python program to implement the Breadth First Search (BFS) algorithm for graph traversal.

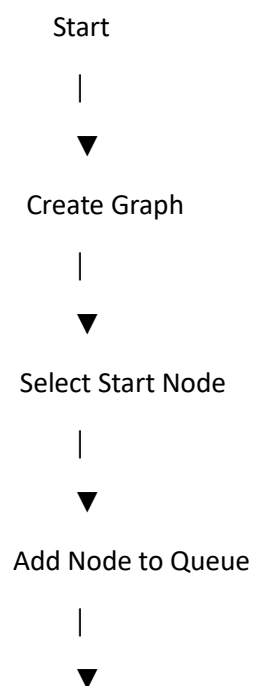
Objective

The objective of this experiment is to understand the Breadth First Search algorithm and implement it to traverse a graph by visiting all nodes level by level.

Algorithm

- Step 1:** Create the graph using an adjacency list.
 - Step 2:** Select the starting node.
 - Step 3:** Initialize a queue and mark the start node as visited.
 - Step 4:** Remove a node from the queue and display it.
 - Step 5:** Add all unvisited adjacent nodes to the queue.
 - Step 6:** Repeat until the queue becomes empty.
 - Step 7:** Stop.
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Flowchart



Queue is Empty?

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No Yes

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Remove Node Stop

Display Node

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▼

Add Unvisited

Adjacent Nodes

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Python Program

```
from collections import deque
```

```
graph = {  
    'A': ['B', 'C'],  
    'B': ['D', 'E'],  
    'C': ['F'],  
    'D': [],  
    'E': ['F'],  
    'F': []  
}
```

```
visited = set()
```

```
queue = deque(['A'])
```

```
visited.add('A')
```

```
print("BFS Traversal:")

while queue:
    node = queue.popleft()
    print(node, end=" ")

    for neighbor in graph[node]:
        if neighbor not in visited:
            visited.add(neighbor)
            queue.append(neighbor)
```

Sample Output

BFS Traversal:
A B C D E F

Result

The graph was successfully traversed using the Breadth First Search algorithm.

Conclusion

Thus, the Breadth First Search (BFS) algorithm successfully visited all the nodes of the graph level by level using a queue.